



Arnab Mandal

Information Science

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Skills

- Analytical Thinking
- Problem-solving
- Communication
- Organization
- Team Collaboration
- Debugging
- Research

Technical Skills

C, C++
Git, GitHub
Linux
Visualization (Power BI)
Python (Pandas, NumPy)
SQL Query Writing
RAG Embeddings
Vector Databases (pgVectors)
Supabase, PostgreSQL
REST APIs

Languages

English - Fluent
Hindi - Fluent
Bengali - Native

Interests

Exploring Complex Systems
Fiction and Literature
SoulsBorne Games

Projects

StudyBuddy – AI Document Assistant (RAG based)

- Built a RAG system for semantic search and AI chat over PDF documents
- Implemented embeddings and cosine similarity retrieval using pgVectors
- Developed citation-based responses with real-time streaming and adaptive quiz system

AI Curriculum builder & Learning Assistant (Team Project)

- Built on AI systems to generate learning roadmaps based on user input
- Designed prompt workflows to ensure coherent, explanatory outputs
- Integrated YouTube API and developed full-stack app with persistent storage

Cerberus – Multi-node Failover System (Team Project)

- Designed a 3-node ESP32 gateway architecture with primary, secondary and connectivity nodes for failover
- Developed redundant communication enabling automatic backup-node takeover during primary failure
- Implemented dual-mode monitoring with internet and offline dashboards via connectivity node's Wi-Fi hotspot

ExamCraft – Timed MCQ & CBT Exam Platform

- Built and deployed a full-stack examination platform with Supabase-backed PDF question-paper storage
- Implemented OMR-based tests, automatic evaluation, and performance dashboards for score tracking
- Enabled users to create and publish custom tests, allowing other users to access and attempt them

Certifications

Programming in C	Springboard, Infosys
Data Science for Engineers	NPTEL
AI Skills Passport	EY & Microsoft

Education

10th CBSE, IIS Asansol	
96%	2021 - 2022
12 th CBSE, IIS Asansol	
93.4%	2022 - 2023
BE in Information Science & Engineering, RVCE	
CGPA: 9.64	2024 - present